FUNCTIONAL DOCUMENT – GLOF MONITORING SYSTEM

1. Introduction

The GLOF Monitoring System is designed to detect, analyze, and generate early warnings for Glacial Lake Outburst Flood (GLOF) hazards using satellite imagery and hydrological modeling.

The system integrates multi-temporal satellite datasets (Sentinel-2 and Landsat-5) with Digital Elevation Models (DEM) and machine learning techniques to automate the identification of glacial lakes and assess their potential risk.

The system transitions traditional manual monitoring into a fully automated computational pipeline capable of generating reliable hazard warnings for disaster management authorities.

2. Product Goal

The primary goal of this system is to:

- Automate the detection and monitoring of glacial lakes
- Analyze long-term glacier changes using historical datasets
- Compute morphometric and hydrological parameters
- Generate a **GLOF Risk Index** for each micro-watershed
- Provide categorized warning levels (Very Low → High Hazard)

This contributes to disaster risk reduction by enabling early warning and decision-making support.

3. Demography (Users, Location)

Users

- Disaster Management Authorities
- Environmental Researchers
- Government Agencies
- Hydrology and Climate Scientists

User Characteristics

- Technical background in GIS, Remote Sensing, or Data Analysis
- Requires high accuracy and reliable outputs
- Uses analytical dashboards and reports

Location

- Target Region: High Mountain Asia (Himalayan Region)
- Focus Area: Glacial regions with potential GLOF risk

4. Business Processes

The system follows the below core processes:

1. Data Acquisition

- Collect satellite data from Sentinel-2 (recent) and Landsat-5 (historical)

2. DEM Processing

- Perform sink filling
- Compute flow direction
- Generate flow accumulation
- Delineate micro-watersheds (413 MWs)

3. Spectral Index Computation

- Calculate NDWI (water detection)
- Calculate NDSI (snow/ice detection)

4. Machine Learning Classification

- Classify terrain into:
  - Glacier
  - Lake
  - Non-glacial land
- Use Decision Tree model

5. Morphometric Analysis

- Compute:
  - AAR (Accumulation Area Ratio)
  - SI (Snout Index)
  - CCI (Catchment Compactness Index)
  - SDI

6. GLOF Index Generation

- Combine all parameters
- Generate 5-level hazard classification

7. Warning System

- Output risk levels:
  - Very Low
  - Low
  - Medium
  - High
  - Very High

5. Features

5.1 Feature #1: Automated Glacial Lake Detection

1. Description

The system automatically detects glacial lakes using spectral indices (NDWI, NDSI) and classifies terrain using machine learning.

2. User Story

As a disaster analyst, I want the system to automatically detect glacial lakes so that I can monitor potential hazards without manual mapping.

5.2 Feature:

1. Description

The system generates micro-watersheds using DEM data and identifies downstream connectivity of glacial lakes.

2. User Story

As a hydrologist, I want to analyze watershed flow paths so that I can understand how floodwater will travel downstream.

5.3 Feature:

1. Description

The system calculates glacier and lake parameters such as AAR, SI, and CCI to assess structural stability and risk.

2. User Story

As a researcher, I want to evaluate glacier properties so that I can determine the likelihood of lake outburst events.

5.4 Feature:

1. Description

The system generates a GLOF Index using Decision Tree classification to categorize risk levels.

2. User Story

As a disaster management officer, I want risk classification outputs so that I can prioritize high-risk areas.

5.5 Feature:

1. Description

The system produces hazard-level warnings for each micro-watershed.

2. User Story

As an authority, I want early warnings so that I can take preventive measures before a disaster occurs.

6. Authorization Matrix

Role	Access Level
Administrator	Full access to system, datasets, and configuration
Researcher	Access to analysis tools and datasets
Analyst	Access to results, reports, and visualization
Viewer	Read-only access to final outputs

7. Assumptions

- Satellite data availability is consistent (Sentinel-2 and Landsat-5)
- DEM data (ASTER GDEM) provides sufficient resolution
- Machine learning model (Decision Tree) performs consistently
- Cloud cover may affect optical data availability
- System operates in a stable computational environment
- Users have basic knowledge of GIS and remote sensing